Solar Radiation: Analysis and Prediction

STAC WARS - PS 2 | Voyage-123 solar observation data

1. Dataset and Preparation

The Voyage-123 file has **32,686 readings** taken every 5 minutes from 1 Sep to 31 Dec 2016 (118 days) with 11 columns: UNIXTime, Data, Time, Radiation, Temperature, Pressure, Humidity, WindDirection, Speed, TimeSunRise and TimeSunSet. There are no missing values and no duplicate rows. The file is stored newest-first, so it was sorted by time. The *Data* column is really the date (its time part is always 12:00:00 AM), so it was combined with *Time* to build a proper datetime column; sunrise and sunset were converted to datetimes the same way. Columns were renamed *WindDirection(Degrees)* to *WindDirection* and *Speed* to *WindSpeed*. No outliers were removed, because the task asks for them to be identified and explained.

2. Part A: Data Visualisation

2.1 Radiation over time

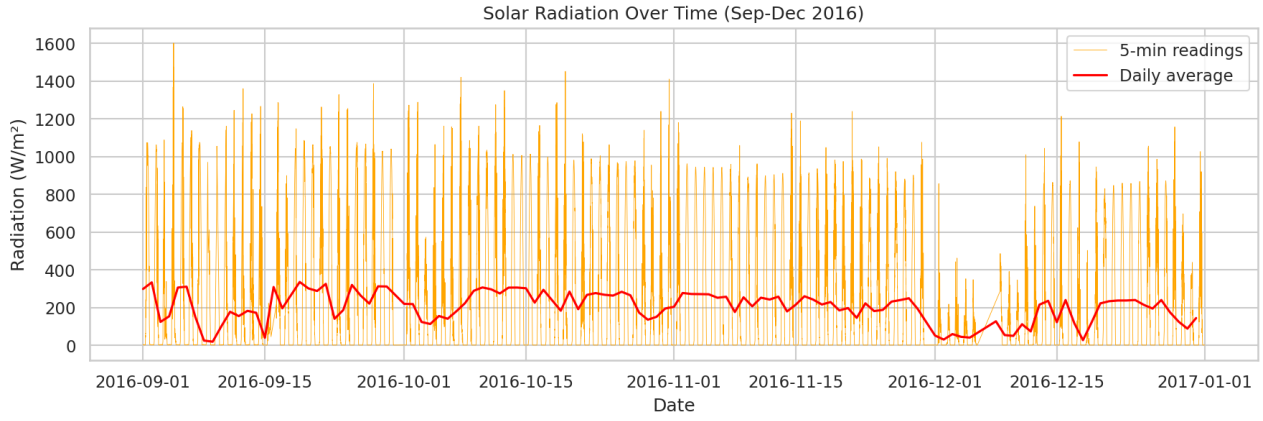


Figure 1. Radiation across the whole period. Orange: 5-minute readings; red: daily average. The gaps (for example early December) are missing data.

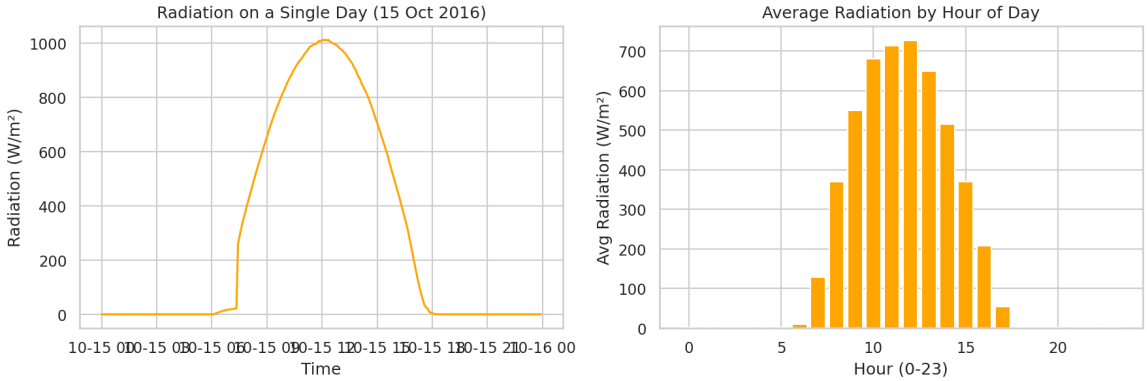


Figure 2. Left: radiation on one day (15 Oct) forms a bell curve. Right: average radiation by hour of day, peaking at noon.

2.2 Radiation vs other environmental parameters

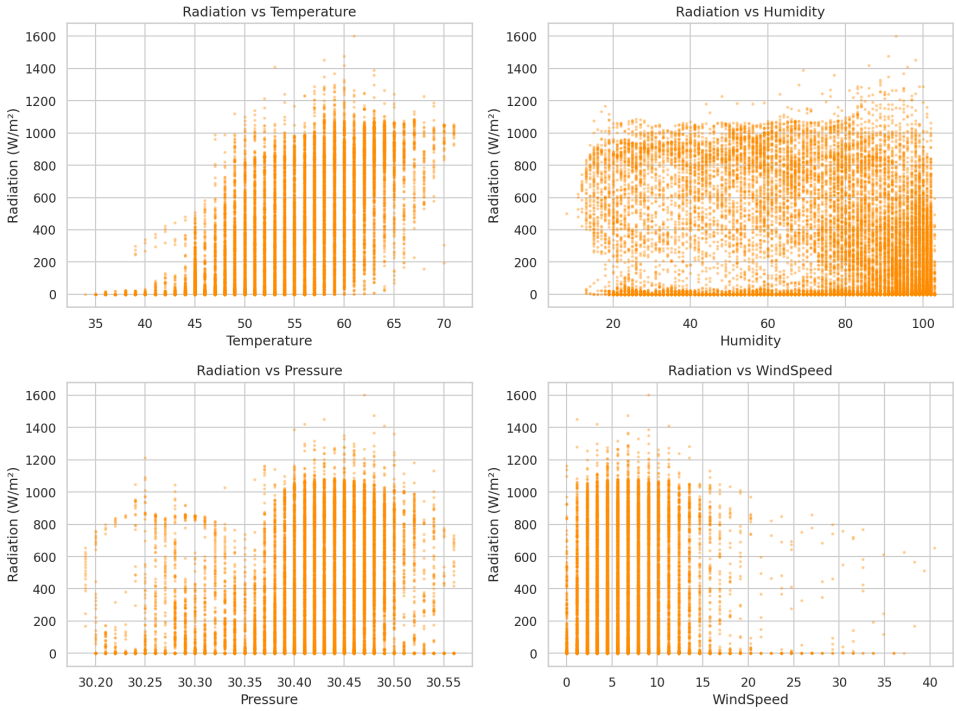


Figure 3. Radiation against temperature, humidity, pressure and wind speed.

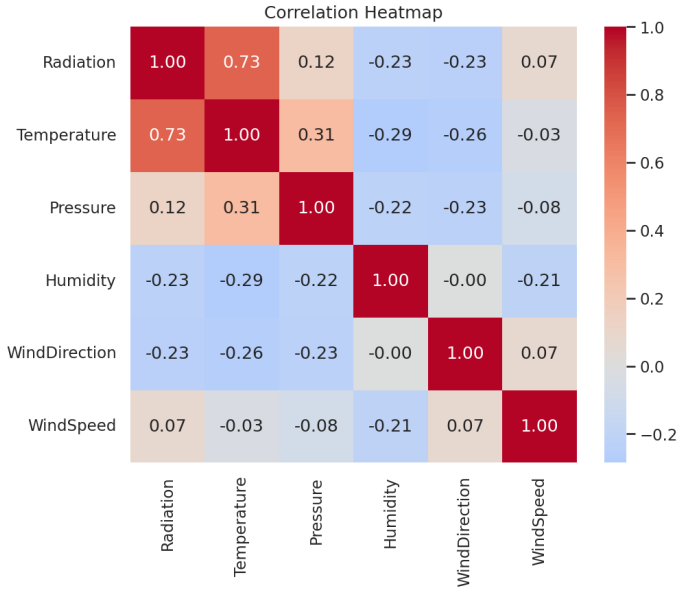


Figure 4. Correlation heatmap. Temperature has the strongest link with radiation (0.73).

2.3 Outliers

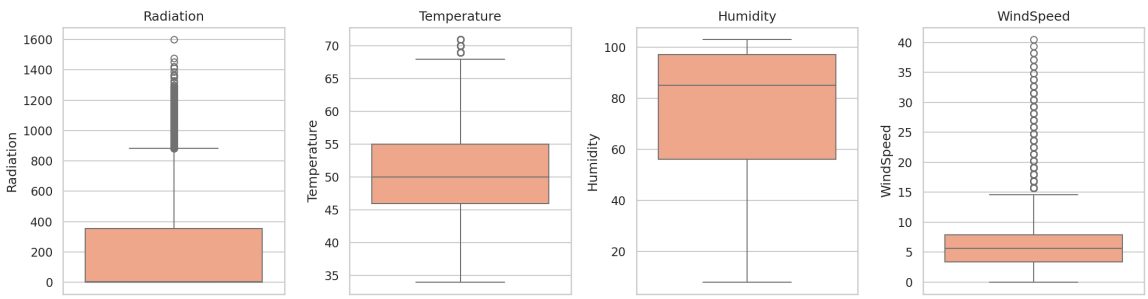


Figure 5. Boxplots of radiation, temperature, humidity and wind speed.

2.4 Bonus visualisation

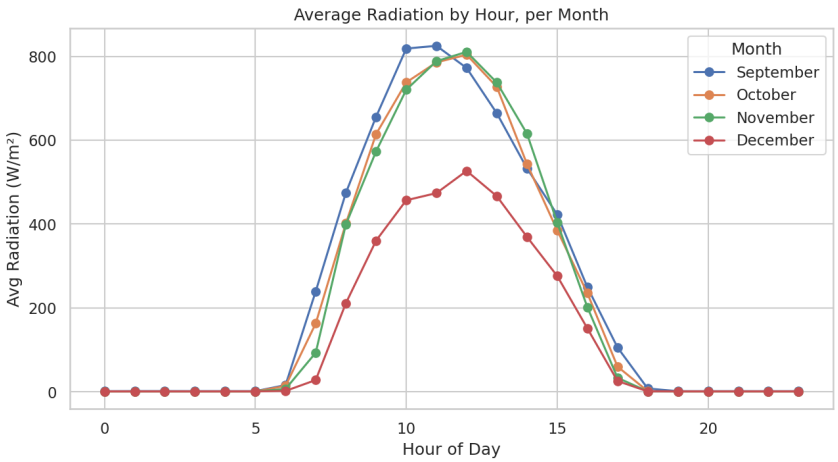


Figure 6. Average radiation by hour for each month. December is much lower than September to November.

3. Part B: Inferences

1. Time of day with the highest radiation. The hourly average peaks at 12:00-12:59 (about 727 W/m^2). Readings from 10:00-14:00 average about 657 W/m^2. Radiation rises sharply from 07:00, falls after 13:00 and is near zero (about 1.4 W/m^2) from 19:00 to 05:59. About 47% of all readings are night-time baseline values (2 W/m^2 or less).

2. Strongest relationship. Temperature (r = 0.73 overall, 0.63 daytime only). Humidity is next and negative (r = -0.23 overall, -0.41 daytime). Pressure is weak (+0.12) and wind speed is almost unrelated (+0.07). Wind direction shows -0.23, but it is an angle (0 and 360 degrees are the same direction), so a straight-line correlation is not reliable for it. Temperature is largely a result of sunlight heating the surface, so this is correlation, not proof of cause.

3. Radiation, temperature and humidity. Temperature follows radiation through the day: both peak around 12:00 (average temperature about 58 at noon vs about 46 at 05:00). Humidity moves the opposite way: lowest in the morning (about 69% at 09:00-11:00) and highest in the late afternoon and evening (about 81%). Temperature and humidity are negatively correlated (r = -0.29). During the day, radiation drops sharply once humidity passes 80%:

Daytime humidity band	0-40%	40-60%	60-80%	80-100%	Above 100%
Avg radiation (W/m^2)	586	663	577	336	216

High humidity usually goes with cloud, fog or rain, which block sunlight.

4. Outliers and unusual observations.

- **Humidity above 100%:** 4,311 readings (about 13%), maximum 103%. Relative humidity cannot physically exceed 100%, so this is likely sensor over-range near saturation (fog or dew). It is more common in the evening and at night.
- **Wind speed:** maximum 40.5 on 16 Dec at 14:45, against a mean of 6.2. About 479 readings fall above the boxplot whisker (14.6). These are likely real storm gusts.
- **Radiation:** the boxplot flags daytime values as outliers only because about half the readings are night-time near-zero. The real anomaly is the 1,601 W/m^2 peak on 4 Sep at 12:15. Only 5 readings exceed 1,400, while typical daytime maxima are about 1,000-1,300. A sensor spike or a cloud-edge reflection effect is the likely cause.
- **Data gaps:** several stretches of missing data, the longest about 2.6 days (5-8 Dec), visible in Figure 1. Only 118 of the 122 calendar days are present.

Bonus insights

- **December drop is bigger than day length explains.** Average radiation is nearly flat from September to November (about 230 W/m^2) even though daylight shrinks from 12.3 h to 11.2 h. In December it falls to about 141 W/m^2 (roughly 39% lower) while daylight shrinks by only about 0.3 h more. Midday radiation in December is also only about 527 W/m^2 vs about 800 earlier. This is likely due to lower sun angle and cloudier weather, plus the December data gaps.
- **The highest reading did not come from the sunniest day.** The 1,601 W/m^2 peak was on a day whose average radiation was only 153.6 W/m^2, while the top days average above 310 W/m^2. Extreme peaks come from short bursts, not from consistently bright days.

4. Part C: Solar Day Analysis

Readings were grouped by date to compute the daily maximum and average radiation (118 days, saved as *daily_solar_report.csv*).

Item	Result
Day with highest radiation	2016-09-04 (1601.26 W/m^2)
Time of peak	12:15:04
Temperature at peak	61
Humidity at peak	93%
Wind speed at peak	9.0
Sunrise / Sunset that day	06:08 / 18:35 (12 h 27 min of daylight)
Peak after sunrise	6 h 07 min 04 s (about 6.12 h), roughly solar noon

Top 5 days by average radiation

Rank	Date	Daily average (W/m^2)	Daily max (W/m^2)
1	2016-09-19	335.50	1085.75
2	2016-09-02	333.47	1062.01
3	2016-09-22	325.52	1053.17
4	2016-09-25	319.92	1075.29
5	2016-09-28	312.40	1029.30

All five best days are in September, and none of them is the day of the record peak (4 Sep).

5. Bonus: Solar Radiation Prediction

Features. Environmental: Temperature, Pressure, Humidity, WindDirection, WindSpeed. Temporal: HourDecimal (hour + minutes/60) and DayOfYear. Temporal features were included because Part B showed radiation depends strongly on time of day. UNIXTime and the sunrise/sunset text columns were excluded.

Split. 80% train (26,148 rows), 20% test (6,538 rows), random_state = 42. **Models.** Linear Regression (baseline) and Random Forest Regressor (100 trees).

Model	RMSE (W/m^2)	R^2
Linear Regression	191.27	0.624
Random Forest	75.81	0.941
Random Forest, train on past / test on future	134.54	0.743

Random Forest beats the linear baseline because radiation follows a bell curve through the day, not a straight line, and the forest captures that non-linear pattern.

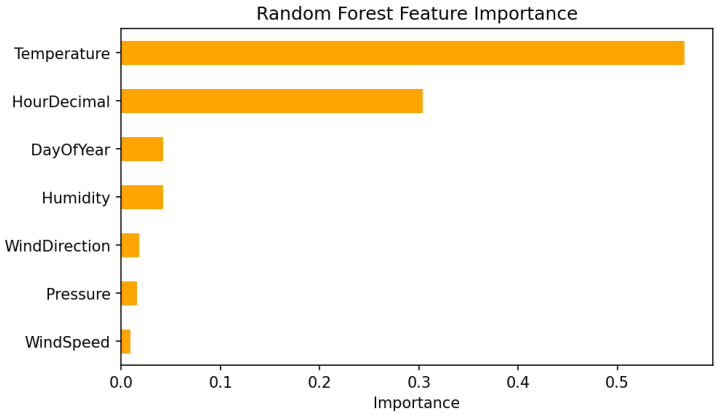


Figure 7. Random Forest feature importance.

Useful features. Temperature (56.8%) and HourDecimal (30.4%) together account for about 87% of the importance. DayOfYear (4.2%) and Humidity (4.2%) add a little. Pressure, WindDirection and WindSpeed contribute under 2% each, which matches Part B, where wind and pressure were only weakly related to radiation.

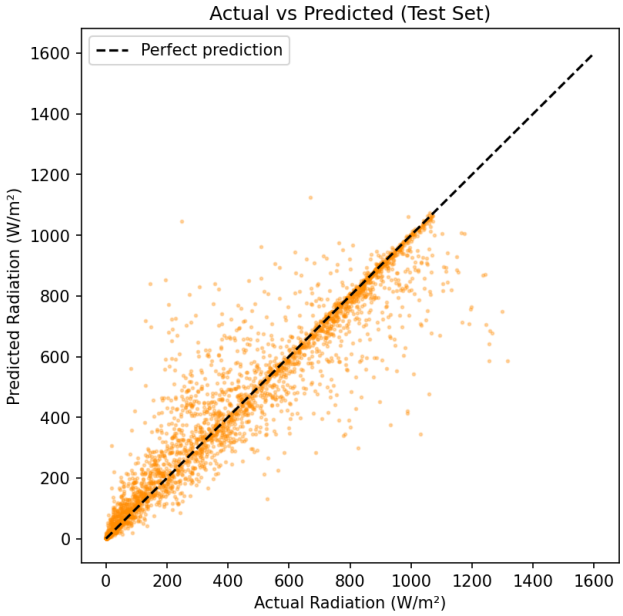


Figure 8. Actual vs predicted radiation on the test set (dashed line = perfect prediction).

Caveat. Readings are every 5 minutes, so neighbours in a random split are near-identical, which makes R^2 = 0.941 optimistic. The stricter test (train on earlier dates, predict later ones) gives R^2 = 0.743, a more realistic estimate for new days. Temperature is also partly an effect of sunlight, so in true forecasting it would not be known in advance.

6. Conclusion

Radiation follows a clear daily cycle peaking at noon, rises with temperature and falls with high humidity. December is markedly weaker than the earlier months. The data has real quirks (humidity above 100%, wind gusts, missing days) that were reported rather than removed. A Random Forest using temperature and time features predicts radiation with RMSE 75.81 W/m^2 and R^2 0.941 on a random test split, and R^2 0.743 when predicting later dates from earlier ones.